

Structure-Aware Global Mixed-Integer Nonlinear Optimisation

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Global Optimisation: Problem Statement

Optimisation problem:

$$\begin{aligned} \min f(\mathbf{x}) & \quad \text{(objective function)} \\ \text{s.t. } g_k(\mathbf{x}) \leq 0 \quad \forall k \in \mathcal{C}, & \quad \text{(constraints)} \\ \mathbf{x}_i \in \mathbb{R} \quad \forall i \in \mathcal{I}_C, \quad \mathbf{x}_i \in \mathbb{Z} \quad \forall i \in \mathcal{I}_I, \quad \mathbf{x}_i \in \{0, 1\} \quad \forall i \in \mathcal{I}_B. & \quad \text{(continuous, integer, binary variables)} \end{aligned}$$

- f, g differentiable
- Constraint qualifications satisfied (no degenerate behaviour of constraint gradients)

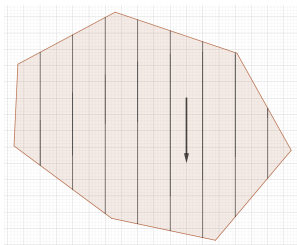
My work is concerned with problems that are **nonlinear** (including **nonconvex**) and/or **mixed-integer**.

Global optimisation seeks global optima (optimal for the entire feasible set) rather than local optima.

Main Challenges

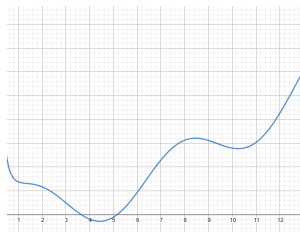
Mixed-integer programs:

- Cannot follow a continuous improving path
- Exponential number of combinations of discrete variable values



Nonconvex nonlinear programs:

- Local optima that are not global
- Difficult to certify infeasibility



How to **prove optimality**, or compute how far from optimal a solution is?

How to **find good feasible solutions** fast?

Optimality Conditions

More commonly employed in continuous optimisation.

First-order conditions:

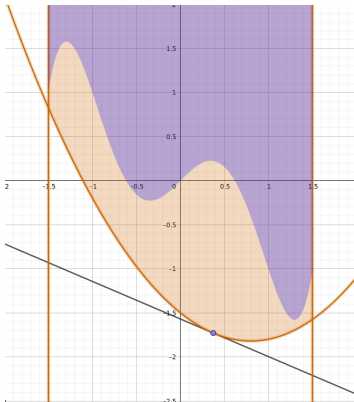
Karush-Kuhn-Tucker conditions are satisfied at a feasible point x^* if there exists a nonnegative multiplier λ such that:

$$\begin{aligned} -\nabla f(x^*) &= \sum_{k \in \mathcal{C}} \lambda_k \nabla g_k(x^*) \\ \lambda_i g_i(x^*) &= 0 \quad \forall i \in \mathcal{C} \end{aligned}$$

KKT conditions are necessary for local optimality and, if the problem is convex, sufficient for global optimality.

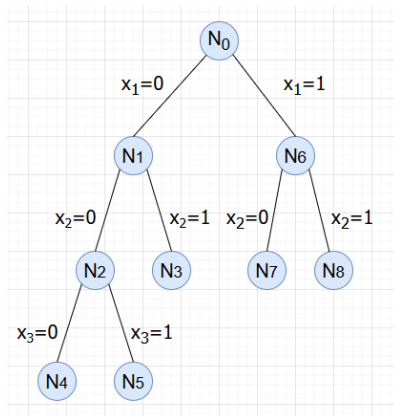
Optimality conditions for various problem classes are an active research area.

Relaxations



- Relaxations are sets containing the feasible set
- Usually convex or linear
- Yield **lower bounds** on the objective
- Tighter relaxation $\Rightarrow$ better bound
- Relaxations are often iteratively strengthened by adding **cutting planes** (valid linear inequalities)

Branch-and-Bound



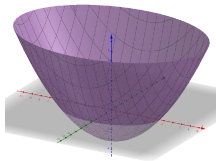
(Figure created with the tool <https://app.diagrams.net>)

- Global optimisation approach for mixed-integer and/or nonlinear nonconvex problems
- Recursively **divides the feasible set**, subproblems form a tree
- Feasible solutions $\rightarrow$ upper bounds, relaxations $\rightarrow$ lower bounds
- Primal heuristics to find good feasible solutions early
- Continues until gap between upper and lower bounds is closed or below tolerance

On Local Optima of Box-Constrained Quadratic Programs

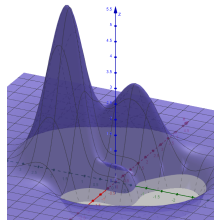
Generalised Convexity: Motivation

- Nonconvexity $\rightarrow$ local optima and non-optimal KKT points, optimality/infeasibility difficult to prove



convex

or



nonconvex

- Some nonconvex problems share certain properties with convex problems
- Empirical example: for some optimal power flow and pooling problems, solvers tend to converge to the global optimum
- Research on generalised convexity is concerned with leveraging such similarities
- **Goal:** tackle nonconvexity by understanding what makes many nonconvex problems challenging

Invex Functions and Problems

Definition

Function f is invex if: $\exists \eta \mid \eta(x, y) \nabla f(x) \leq f(y) - f(x)$ for all $x, y \in X$
(Hanson, 1981)

Martin, 1985: relaxed Hanson's conditions, definition of KKT-invex problems:

Definition

An optimisation problem is KKT-invex if there exists a function η such that:

$$g(x) \leq 0, g(u) \leq 0 \Rightarrow \begin{cases} f(x) - f(u) - \nabla f(u) \cdot \eta(x, u) \geq 0, \\ \text{for } i \in \mathcal{C}, \text{ if } g_i(u) = 0, \text{ then } -\nabla g_i(u) \cdot \eta(x, u) \geq 0. \end{cases}$$

Every KKT point is a global optimum if and only if the problem is KKT-invex.

Boundary-Invexity

New approach to KKT-invexity, based on a special subclass of stationary points of subproblems that indicate a violation of KKT-invexity.

(Bestuzheva, Hijazi, 2019)

- Boundary-invexity is sufficient for KKT-invexity of problems in $\mathbb{R}^2$ and a weaker version of boundary-invexity is necessary for KKT-invexity of problems in $\mathbb{R}^n$
- **Can be verified algorithmically**
- Used to show that optimal power problems for one line with realistic parameters are KKT-invex

Further Shift of Perspective: Optima-Invexity

Definition

A problem is optima-invex if it has a unique connected locally optimal subset or a unique connected set of unbounded directions.

KKT-invexity	Optima-invexity
Allows multiple basins of attraction (if optima are equivalent)	Does not allow multiple basins of attraction
Does not allow non-optimal KKT points	Allows some non-optimal KKT points

- Focus on local optima/unbounded directions and their basins of attraction
 - In practice, convergence to non-optimal KKT points tends to not be an issue (Lee et al., 2016)
 - Optima-invexity serves as a basis for studying the relation between non-optimal KKT points and local optima

Conditions for Optima-Invexity

Conjecture

If an optimisation problem does not have non-optimal KKT points, then it is optima-invex.

That is, the absence of non-optimal KKT points implies the absence of multiple distinct local optima or sets of unbounded directions.

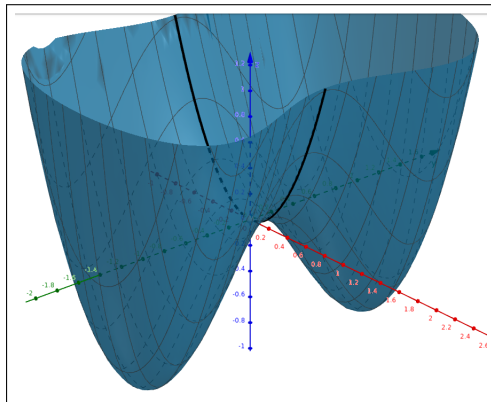
Proof Approach - Unconstrained Case

Step 1: Assume the contrary: there exist distinct local optima $\mathbf{x}^{1*}$, $\mathbf{x}^{2*}$.

Step 2: consider basins of attraction A_1 , A_2 of $\mathbf{x}^{1*}$, $\mathbf{x}^{2*}$ respectively, and the shared boundary δA .
Prove that $\delta A \not\subset A_i$ for $i = 1, 2$.

Step 3: consider $x^B \in \delta A$. Two possibilities:

- x^B is a non-optimal stationary point or
- x^B belongs to a basin of attraction of a non-optimal stationary point.

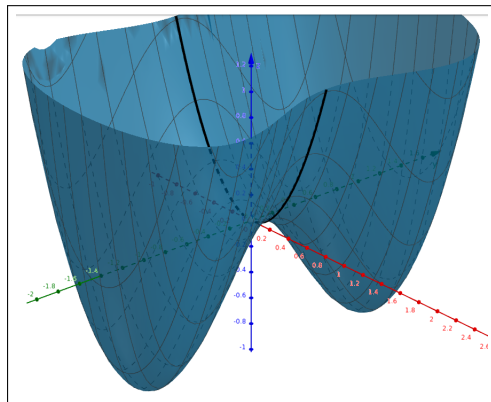


Ridges

Definition

A **ridge** is a hypersurface separating different basins of attraction w.r.t. a given algorithm.

Important for algorithms: ridges tell us how to partition spaces.



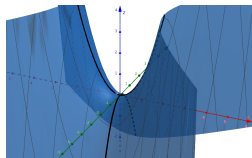
Optima-Invexity of Unconstrained Quadratic Problems

Nonconvex quadratic function (that has been diagonalised): $f(x) = \lambda_1 x_1^2 + \lambda_2 x_2^2$.

$$\text{Unbounded directions of } f: \mathcal{U} = \left\{ x \mid \sum_{i \in \mathcal{I}^-} \lambda_i x_i^2 + \sum_{i \in \mathcal{I}^+} \lambda_i x_i^2 < 0 \right\}$$

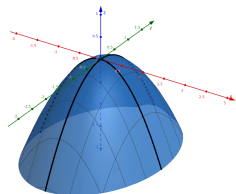
Case 1: one negative eigenvalue $\Rightarrow$ not optima-invex.

- $\mathcal{S}^+$: $(n-1)$ -dimensional subspace spanned by e_i , $i \in \mathcal{I}^+$
- $\mathcal{U} \cap \mathcal{S}^+ = \emptyset$, therefore $\mathcal{U}$ contains two disjoint cones



Case 2: more than one negative eigenvalue $\Rightarrow$ optima-invex.

- Consider the subspace $\mathcal{S}^-$ spanned by e_i , $i \in \mathcal{I}^-$
- There is always a continuous path between any $x^{(1)}, x^{(2)} \in \mathcal{U}$
 - Project into $\mathcal{S}^- \rightarrow$ rotate in $\mathcal{S}^- \rightarrow$ lift

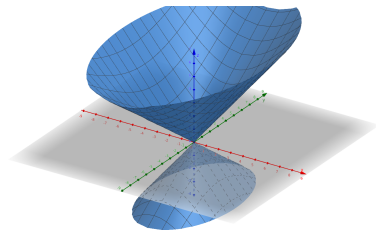


Unconstrained Case: Examples in $\mathbb{R}^3$

$$f(x_1, x_2, x_3) = \lambda_1 x_1^2 + \lambda_2 x_2^2 + \lambda_3 x_3^2$$

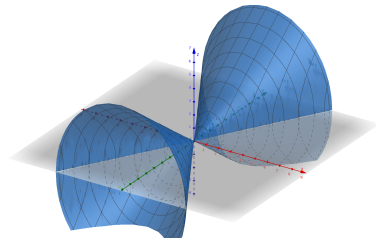
Case 1: one negative eigenvalue ($\lambda_3 < 0$).

- Unbounded rays lie in the two disjoint cones
- $\mathcal{S}^+$ is the (x_1, x_2) -plane
- Not optima-invex



Case 2: more than one negative eigenvalue ($\lambda_1, \lambda_3 < 0$).

- Unbounded rays outside of the cone
- $\mathcal{S}^-$ is the (x_1, x_3) -plane
- Optima-invex



Optima-Invexity of Box-Constrained Concave Quadratic Problems

Consider the problem: $\min f(x)$ s.t. $x \in \mathcal{B}$, where $\mathcal{B} = \underline{x} \leq x \leq \bar{x}$.

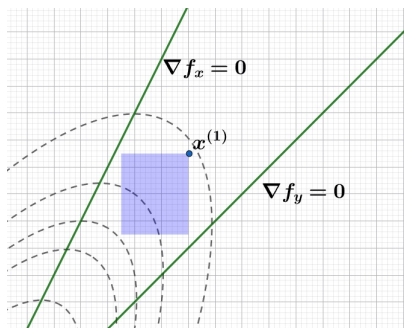
Let $p_i = \{x \mid \nabla f_i(x) = 0\}$

Optima-invexity sufficient condition 1:

if $\mathcal{B} \cap p_i = \emptyset \ \forall i$, then the problem is optima-invex.

Proof outline: use the fact that f increases in the feasible direction along all edges at any local optimum.

Algorithmic use: can be applied directly to prove optima-invexity of some problems.

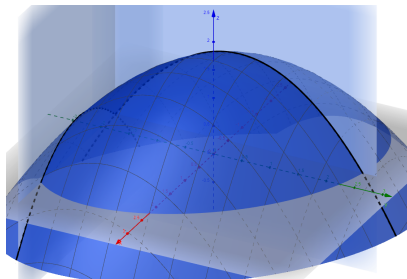


Optima-Invexity of Box-Constrained Concave Quadratic Problems

Optima-invexity sufficient condition 2:

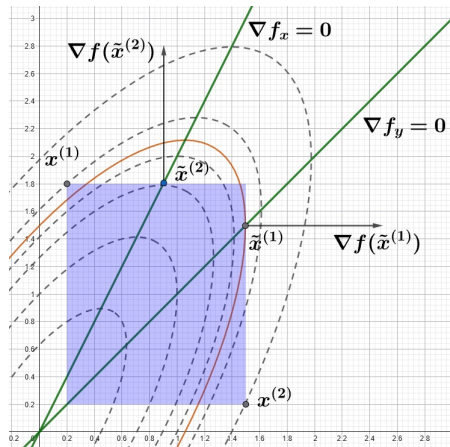
The problem is optima-invex if the following conditions hold:

- 1) for each i, j either
 - a) $\text{int}(\mathcal{B}) \cap p_i \cap p_j = \emptyset$ or
 - b) $\mathcal{B} \cap p_i \cap p_j \neq \emptyset$ for each feasible fixing of x_k , $k \neq i, j$
- 2) the edges of $\mathcal{B}$ do not contain non-optimal KKT points.



Sufficient Condition 2: Proof Outline

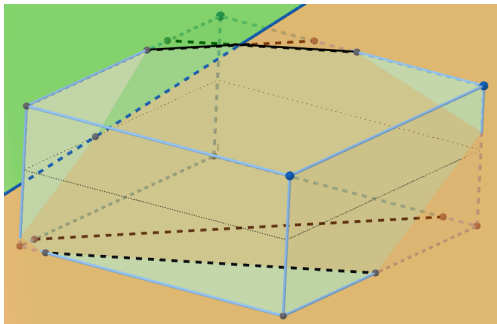
- Local optima x^{*1} and x^{*2} : vertices of $\mathcal{B}$ due to concavity
- Suppose that $x_i^{*1} \neq x_i^{*2} \forall i = 1, \dots, m$
- Consider a path between x^{*1} and x^{*2} along edges $e_1, \dots, e_m$
- The path must cross all $p_1, \dots, p_m$
- Choose the order of $e_1, \dots, e_m$ so that it matches the order of crossing $p_1, \dots, p_m$ (using (1a) and (1b))
- An edge e_k exists that crosses p_k : the crossing point is KKT



(Note: each p_i is crossed an odd number of times. The question is: is p_i crossed an even or odd number of times before p_j is first crossed?)

Algorithmic Use of Sufficient Condition 2

- More expensive to directly apply than sufficient condition 1
- Provides guidelines on how to divide $\mathcal{B}$ to direct the search towards new local optima
- To check property 1: solve linear systems or programs
- To check property 2: solve nonlinear programming subproblems
- If property 1 is violated, partition the domain (e.g. in a branch-and-bound approach)



Practical Uses of Optima-Invexity

- Optimality proofs
 - Challenging in general, may be feasible in special cases
 - Optimality in subproblems: e.g. node subproblem in branch-and-bound / subproblem in a decomposition
- Heuristics (standalone or within branch-and-bound)
 - Use invexity-breaking points to guide the search towards new local optima
 - Can be applied to problems for which there is yet no formal proof of sufficiency

Constraint-Wise Algorithm

First heuristic prototype:

- Rather than compute non-optimal KKT points, consider each relevant constraint and variable separately
- Solve computationally cheap subproblems
- Use solutions to construct a grid
- Begin a new local search in each cell of the grid

Computationally fast and easy to implement, but may result in suboptimal grids.

Example Problem

ex14.1.1 from MINLPLib

(from Floudas et al., Handbook of Test Problems in Local and Global Optimization)

Formulation:

$\min z$ s.t.

$$2y^2 + 4xy - 42x + 4x^3 - 14 \leq z$$

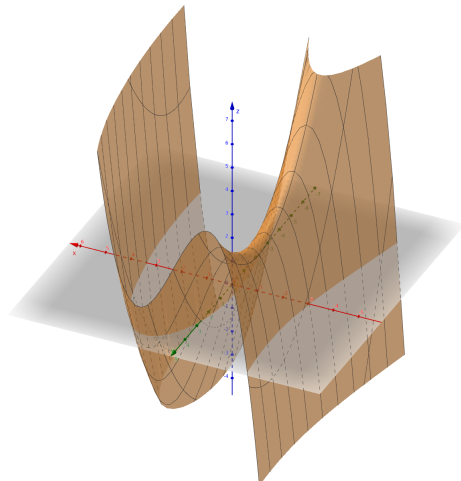
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$$(x, y) \in [-5, 5] \times [-5, 5]$$

Multiple local optima (with the same value)



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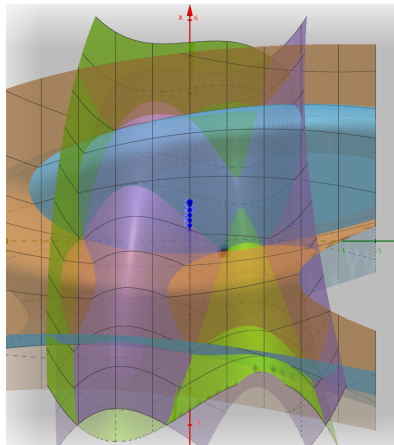
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Algorithm Applied to Example

Grid: $[-5, 5] \times [-5, 5]$

Starting point: $(0,0,5)$

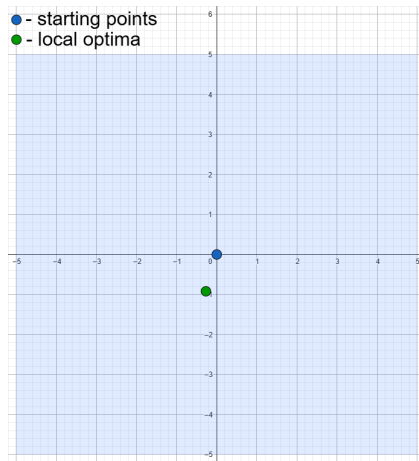
NLP solution: $(-0.27, -0.92, 0)$

Active constraints: (1), (2), (3), (4)

Constraint (1):

- Convex in y , nonconvex in $x \Rightarrow$ focus on x
- Fix $y = 5$, find local optimum (x^*, z^*) of:
 $\max z$ s.t. $g_1(x, 5, z) = 0$
- Add separation line: $x = x^* = -1.36$

Similarly find separation lines for remaining constraints.



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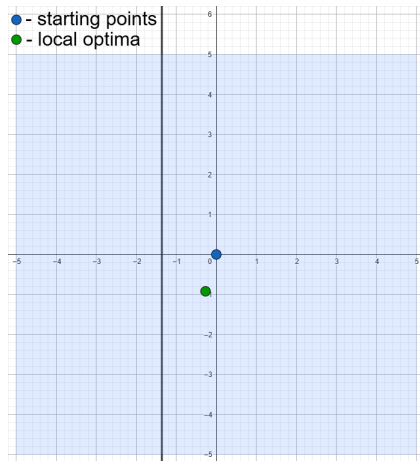
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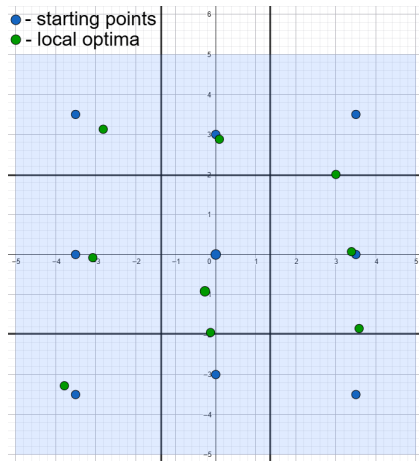
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Similarly find separation lines for remaining constraints.

Result: 9 cells, 9 local optima found.



Summary and Conclusions

- **New approach to nonconvex optimisation:** understand when local optima occur $\rightarrow$ develop efficient algorithms
- **New notion:** optima-invexity
- **Conjecture:** absence of non-optimal KKT points $\Rightarrow$ optima-invexity
- **Sufficient conditions** for optima-invexity of unconstrained quadratic programs
- **Sufficient conditions** for optima-invexity of box-constrained concave quadratic programs
- Invexity-guided **heuristic algorithm** prototype

Research Plan

Research Plan Overview

Main idea: combine insights on *structures* with understanding of *algorithms* to develop *high-performance global optimisation* algorithms.

	Optimality proofs	Strategies within branch-and-bound
Generalised convexity	Conditions	Structure-driven heuristics
Cut impact	Stronger relaxations	Decisions on which cuts to use
Polynomial optimisation	Specialised relaxations	Integration with branch-and-bound
Applications in energy systems, optimal control, telecommunications, social science...		

External collaborations on Dantzig-Wolfe decompositions, reformulation-linearisation technique, optimisation for groundwater simulations

Structure-Aware Nonconvex Optimisation

Optimality conditions	Algorithms
• Study links between non-optimal KKT points and optima-invexity for further problem classes • Refine optima-invexity conditions	• Study convergence of algorithms to local optima • Develop invexity-guided heuristics

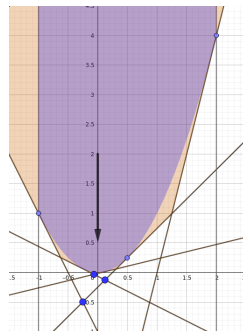
Longer-term: decomposition schemes and multi-level problems with optima-invex subproblems.

Application: social welfare optimisation

- Bilinearly constrained concave quadratic under certain assumptions (Papageorgiou et al., 2023)
- Similar in structure to pooling
- Multi-level $\rightarrow$ KKT conditions (and their implications) play an important role

Cut Impact Analysis

- Cuts are crucial for the performance of branch-and-bound
- Cut management important in decomposition approaches
- **Challenge:** many cuts generated in the solving process → difficult to select which cuts to apply
 - Factors: cut strength, LP relaxation solving cost, side effects
 - Applying the wrong cuts can lead to slowdowns

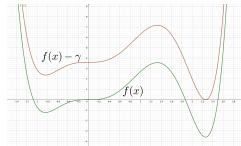


Goal: study the effects of cuts & cut families under realistic assumptions → use the information to enhance branch-and-cut

Polynomial Optimisation

Polynomial optimisation $\Leftrightarrow$ finding the best nonnegative polynomial:

$$\min f(\mathbf{x}) \text{ s.t. } \mathbf{x} \in F \Leftrightarrow \max \gamma \text{ s.t. } f(\mathbf{x}) - \gamma \geq 0 \forall \mathbf{x} \in F.$$



- Decompose f as a sum of nonnegative polynomials $\rightarrow$ nonnegativity certificate
- Certificate-based relaxations can yield strong bounds $\rightarrow$ potential for use within branch-and-bound
- Well-suited for handling infinite constraints ($g(x, w) \leq 0 \forall w \in W$)
- **Challenge:** branch-and-bound tailored to conventional relaxations

Approach: create an integrated certificate-based branch-and-bound algorithm.

- Branching strategies & other components that improve certificate-based relaxations
- Warm-starting techniques
- Adaptive choice of certificate

Integration with LSE Research

LSE research	My research
Within the Department	
Integer and combinatorial optimisation	Cut impact analysis, MINLP topics
Bilevel optimisation, games	Optima-invexity, cut management
Theory of computation, statistics	Algorithm analysis, abstract models
Machine learning	Optima-invexity, algorithm analysis, polynomial certificate selection
Applications: energy, telecommunications, networks	
Broader connections	
Environmental economics	Optimisation for energy systems, integrated problems
Social sciences	Social welfare optimisation
Impact of mathematical and algorithmic developments on environment and society	

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All figures were created by myself with the use of the tool (unless specified otherwise) Geogebra.